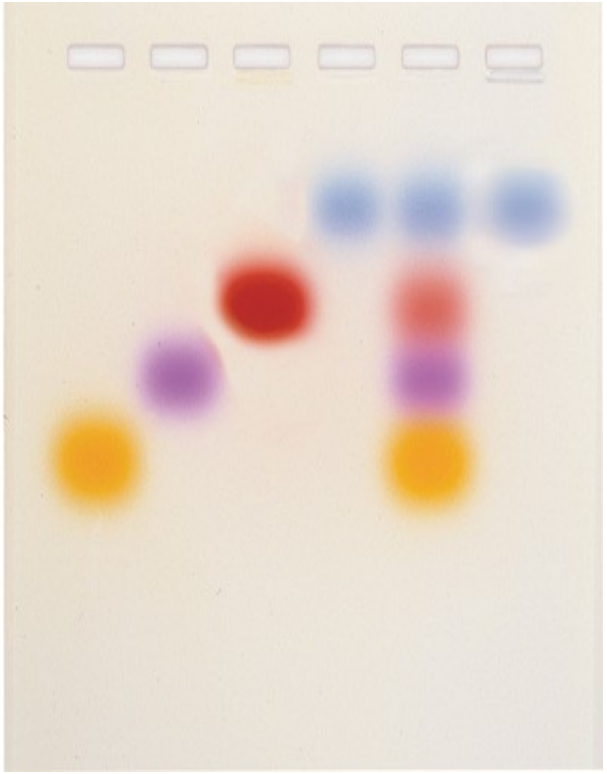
Gel Electrophoresis

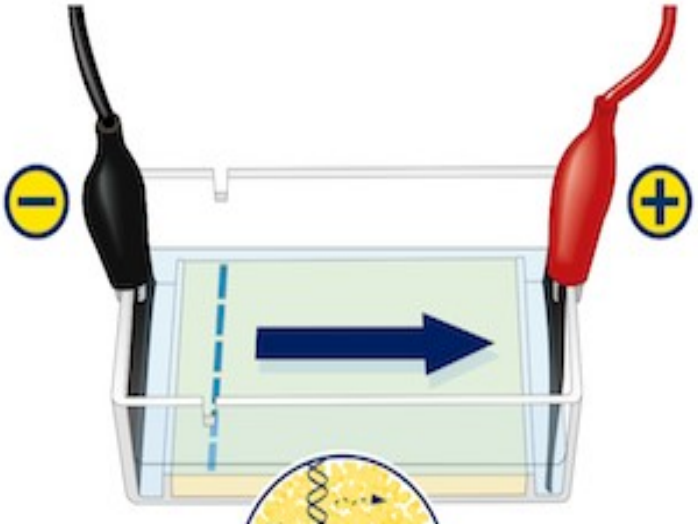


Background and instructions

Gel Electrophoresis: A Separation Process

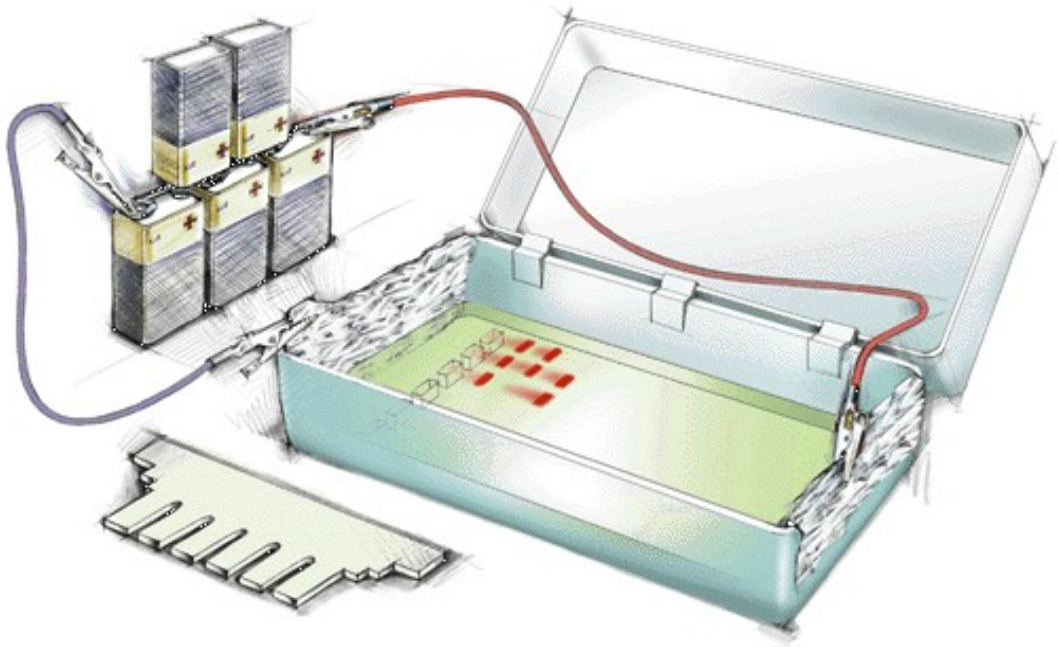
- Gel electrophoresis is a powerful technique used routinely in the areas of human biology, biochemistry, forensics and medicine.
- It separates mixtures based on differences in molecular weight (size), structure and charge of the different component molecules.
- It is commonly applied to the analysis of DNA and proteins.

Negatively charged DNA loaded into the gel at negative electrode



Molecules migrate to positive electrode

Gel electrophoresis in the school laboratory



Electrophoresis apparatus assembled in the lab

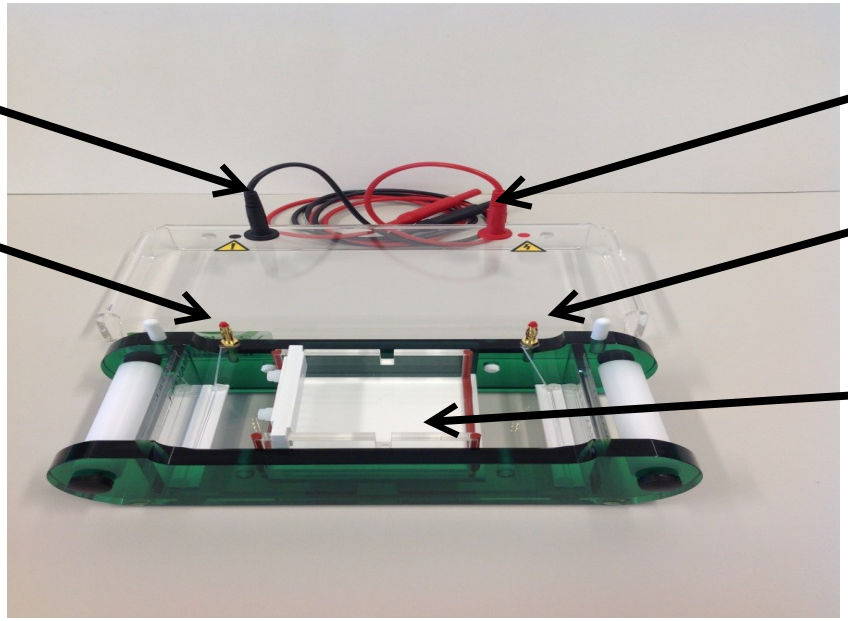
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Gel electrophoresis loan pool sets

What's in it?

- 8 tanks, 8 combs and 8 pairs of leads
- 4 power packs
- 8 fixed volume micropipettes
- 1 variable micropipette
- Dyes
- Pipetting tips
- Eppendorf tubes for dye mixtures
- Resources

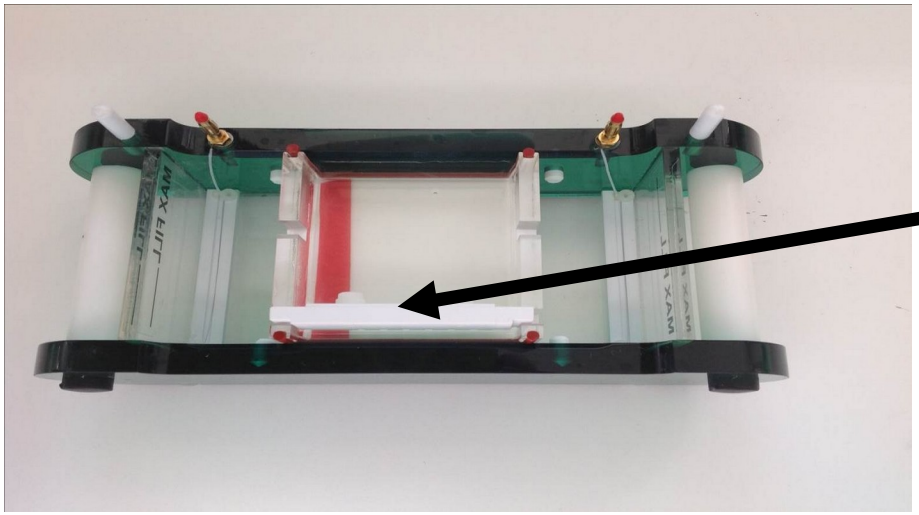
Gel electrophoresis



Commercial electrophoresis equipment
is available in the loan pool

Preparing the gel casting tray

- Rotate the tray so the rubber seals fit in tight against the tank wall
- Place the comb into the edge slot checking the orientation is correct



Comb position and orientation – teeth away from edge

Buffer solution

- The buffer is used to ***create the gel and to fill*** the electrophoresis tank.
- The buffer in the tank helps maintain the correct pH to perform electrophoresis.
- For each tank make a 0.1% (w/v) buffer solution by dissolving 0.5g of sodium hydrogen carbonate in 500mL of tap water.

Making the agar gel

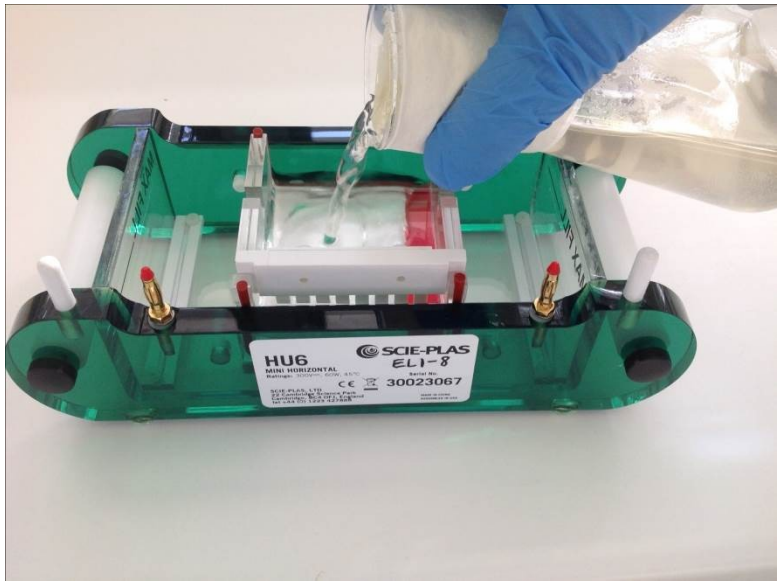
- The gel is the medium along which the dyes will travel and separate out on

- Each tank requires a 1% (w/v) agar solution ie. 1g of agar into 100mL of buffer solution in a flask

- Microwave the flask in short bursts 60 seconds initially, then in 30 second bursts, swirling gently in between until agar solution is clear.

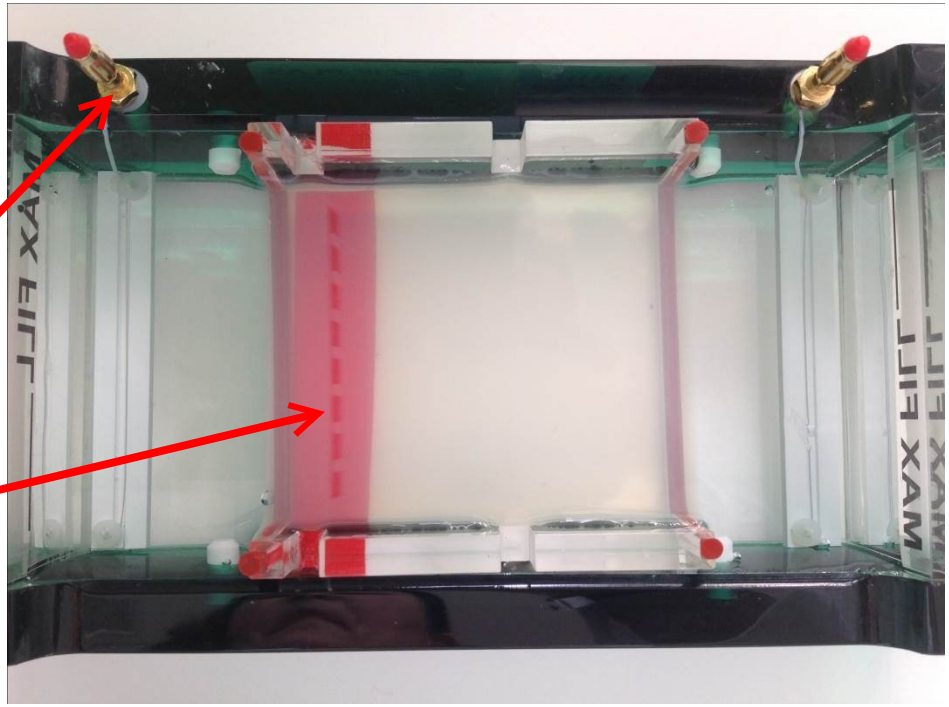
Pouring the agar gel

- The agar must be cooled down to around 60°C before pouring – or it may leak through the seals
- Paper towel around the flask neck enables you to handle the hot flask
- Pour agar to just under the base of the comb slot



Setting up the tank

- Once the gel is set, remove the comb and then lift tray out and rotate it 90 degrees, so that the wells are closest to the negative terminal.

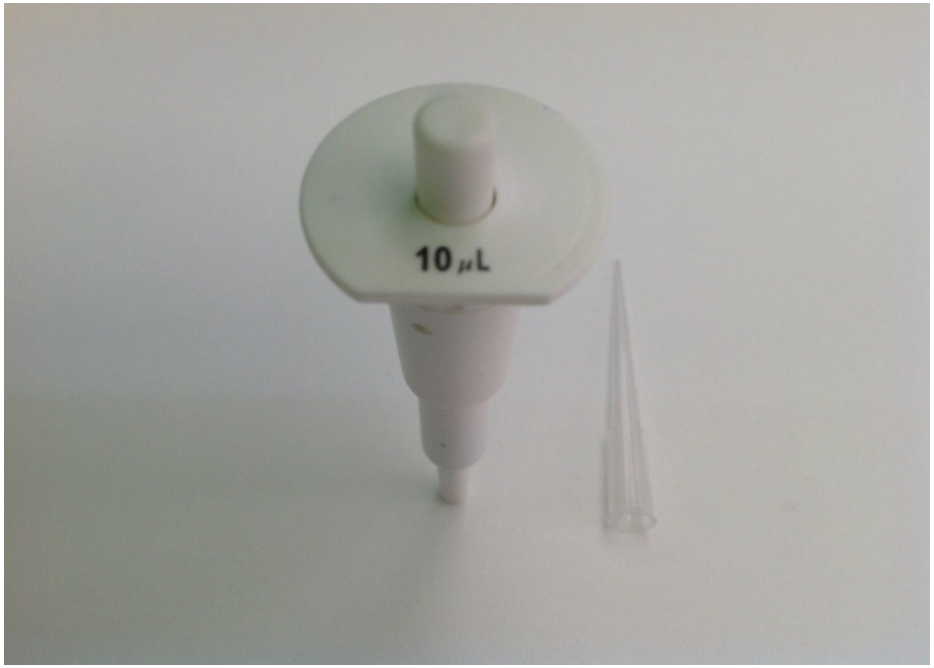


Adding the buffer solution

- Pour the buffer solution slowly into the tank, fill to the ‘max fill’ line – it should just cover the top of the gel
- If the gel starts to float, gently push it back down into place

Fixed volume micropipette

- Micropipette delivers 10uL of dye sample
- Students should practice using these before loading dyes into the gel



Dyes

- The dyes supplied in the loan pool sets are:
  - Bromophenol blue (B)
  - Orange G (OG)
  - Rose Bengal (RB)
  - Tartrazine (T)
  - Xylene cyanol (XC)
  - A ladder, or combination of all dyes (L)
  - Samples (mixtures) numbered 1 through to 4

Adding dyes to the gel

- Select the dyes you want to use or, use the dyes the teacher has supplied
- Write down what you're going to use and in what order *before* pipetting, *see next slide*
- Use a fixed volume micropipette to carefully transfer (or load) 10uL of each dye sample directly into the wells.

Recording your dyes

- Write down which dyes you are going to use, for example:

Well No.	Dye	Results
1	Bromophenol blue	
2	Orange G	
3	Xylene cyanol	
4	Tartrazine	
5	Rose bengal	
6	Ladder	
7	Sample 2	
8	Sample 3	

Negative terminal → Dye separates out → Positive terminal

Safety

Precautions to prevent electrical shock and using electrophoresis apparatus safely include:

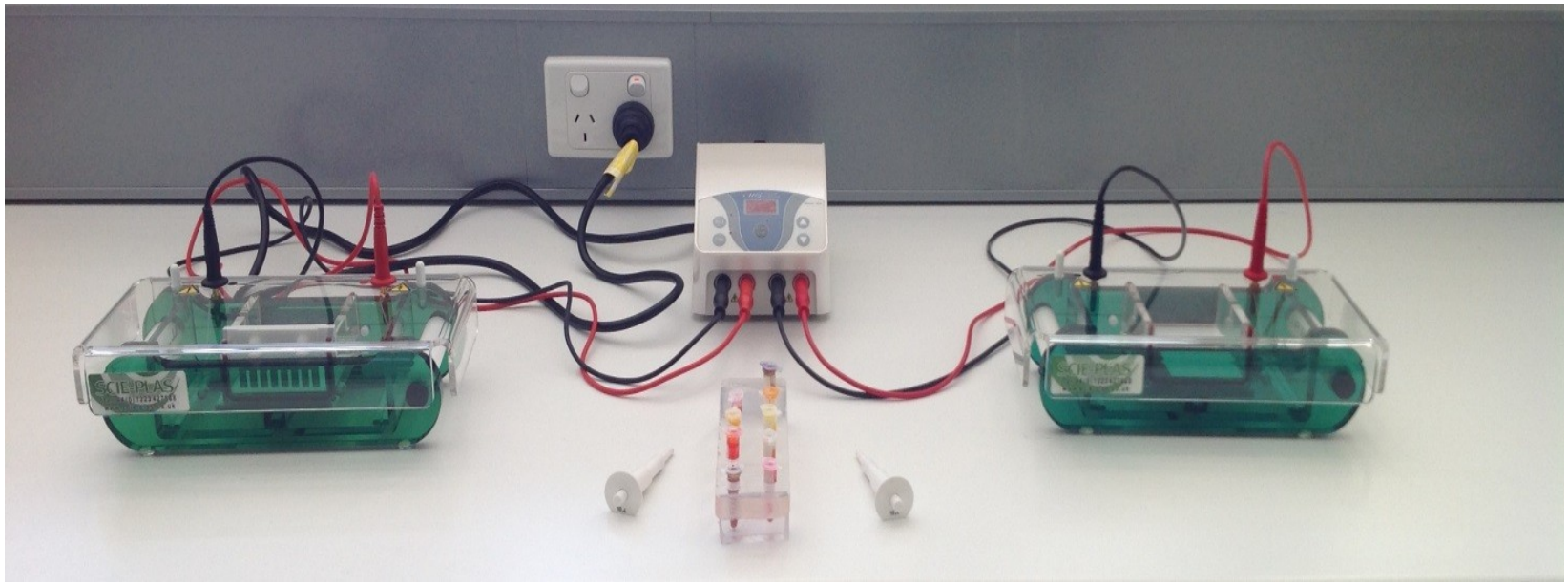
- Ensure power is off before connecting the electrical leads
- Ensure that your hands are dry while connecting the leads
- Keep the apparatus away from sinks or other water sources
- Turn off power before opening lid or reaching inside chamber
- Do not run electrophoresis equipment unattended

Safety

- **Before attaching the lids to the tanks,** ensure that the power packs are NOT connected to power
- Once the gels are loaded with dyes, attach the lids to the tanks
- Attach the leads of the electrophoresis tank to the power pack (red to red and black to black)
- **DO NOT** switch the power pack on until your teacher has checked your set up

Workspace setup

- Tanks set up and connected
- A workstation has two tanks run off of one power pack



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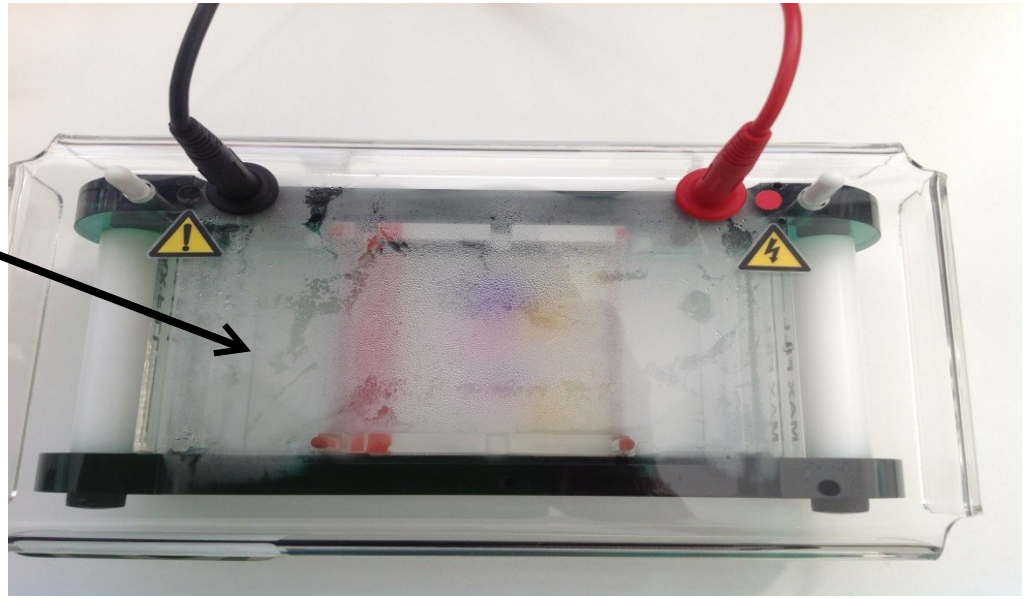
Performing gel electrophoresis

- Once the power pack is connected to the electrophoresis tank, switch on the power pack and set to 100V for 30 minutes
- If the power pack has current adjustment, ensure it is set at 250mA
- Start electrophoresis, bubbles will form at the electrodes of the tank when the voltage is applied - if bubbles are not visible switch the power pack off and notify your teacher
- Press the ‘stop’ button once the power pack has run for 30 minutes

Performing gel electrophoresis

- If left on for too long, dyes will migrate out the end of the gel and the gel could melt
- Fogging up inside the tank during electrophoresis is normal

Condensation forming on the lid



Performing gel electrophoresis

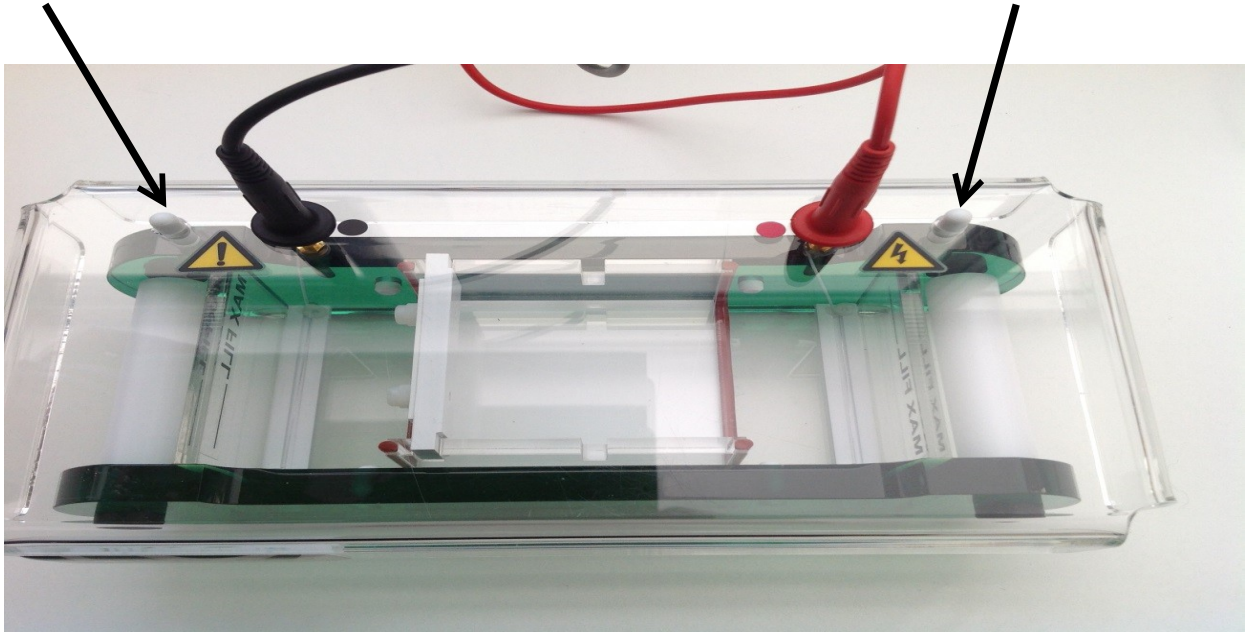
- Once the dyes have migrated sufficiently, turn off the power pack, disconnect it from the power socket, remove leads from power pack and then remove lid by pressing down on white lugs

- Check if the buffer is cool enough, then remove the casting tray with gel inside

- Compare the migration of the sample dyes with the known dyes

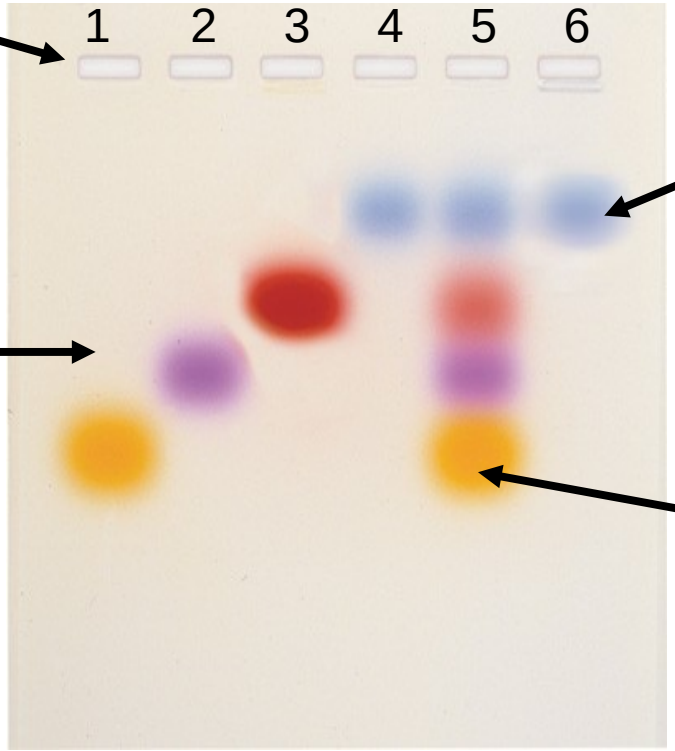
Removing the tank lid

- Lid can be removed by pressing down on white plastic lugs and pulling lid up underneath



Interpreting Results - example separation

Starting point at well



Unknown dye in well No.6 had the same dye as well No.4

Wells 1 – 4 have known dyes

Ladder at well No. 5